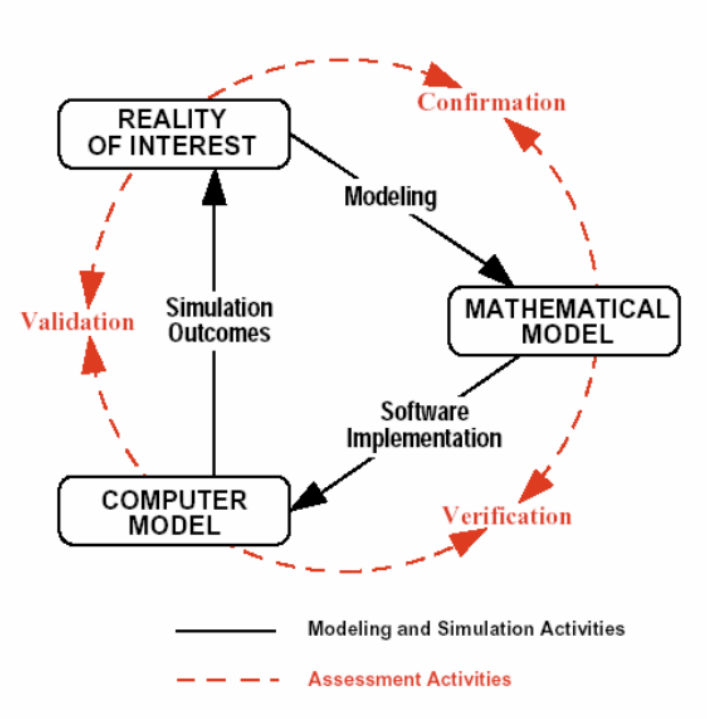
Modelling 2026

Model validation

4 November 2026



From Thackner (see book)

Verification update: See the latest version of Chapter 8 for a verification road map

8.1 Verification

Verification is about making sure your model has been implemented correctly. So how exactly do you do that? Here is a general road map.

1. Review the underlying conceptual and mathematical models. Effective verification without an understanding of these parts is not possible.
2. Ensure that the Python model function(s) docstring(s) is complete for both inputs and outputs, including units for any variables that represent quantities.
3. Check equations for: A. Completeness, with all the terms from the mathematical model, B. conceptual correctness (e.g., flux is equal to a transfer coefficient times a difference), including transport direction (e.g., a surface hotter than air should lose heat energy), and C. unit correctness, by multiplying and dividing units for each equation and confirming they match the expected units for the variable.
4. Check model behavior through careful model function calls. Check results for: A. Reasonable magnitudes, which may be difficult without experience, but alternative models or measurements can help, B. expected behavior at extremes or edge cases, and C. expected direction of effects.

For 3, which equations should you check? Ultimately all of them need to be checked. But it can be helpful to start with the GE(s) and work backwards. You may find that the GE is not written the same way in the code and in the mathematical model. The best pratice in this case is to change one so they match. It is also

You might like verification if you are a perfectionist and excel at finding mistakes

You might like validation if you are interested in real world applications of your model

But if you are a model developer, you need to do both!

Plan for today

1. Verification solution. Volunteers? Questions?
2. Validation concepts. Highlights from book section 8.2.
3. Work on validation exercise.
4. Exercise check-in before the end of class.